

What kind of system is it?

Mathew Smith & Aneta Stefanovska

Consolidating Practical 1 and preparing for Practical 2



Where we are going today

Part A: recap of the methods

What you used in the practical, how the methods compare, and **when to use which**.

- Euler $\rightarrow$ RK2 $\rightarrow$ RK3 $\rightarrow$ RK4
- Newton–Raphson for root finding
- How to check a numerical result

Part B: what kind of system?

Before you simulate or analyse data, form a **hypothesis** about what to expect.

- Isolated, closed, open
- Linear, nonlinear
- Deterministic, stochastic

Part A tells you how to **compute**; Part B uses it to ask what you should **expect**.

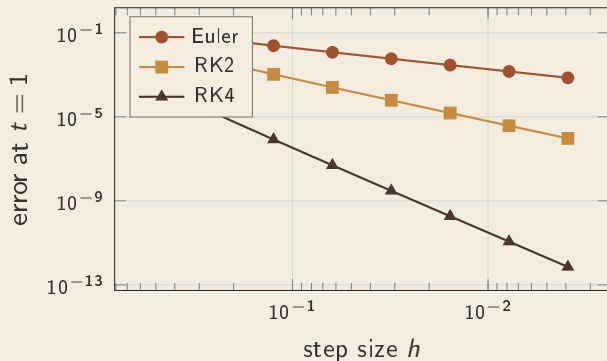
The methods at a glance

Method	Cost per step	Global error	Use when
<i>Time stepping: solving $\dot{y} = f(t, y)$</i>			
Euler (RK1)	1 slope	$O(h)$	Understanding the idea; quick checks
RK2 (midpoint)	2 slopes	$O(h^2)$	A cheap upgrade; rarely the final choice
RK3	3 slopes	$O(h^3)$	A stepping stone to RK4
RK4	4 slopes	$O(h^4)$	Default for smooth, non-stiff problems
<i>Root finding: solving $f(x) = 0$</i>			
Newton–Raphson	f, f'	quadratic	Nonlinear algebraic equations (Kepler)

Two different jobs

Euler and Runge–Kutta advance a solution **in time**. Newton–Raphson finds a number that satisfies an **equation**.

What the methods do: error against step size



Test problem: $\dot{y} = -y$, $y(0) = 1$,
exact $y(1) = e^{-1}$.

On a log-log plot the error is a straight line, and its **slope is the order** of the method.

Read the plot

Same h , very different accuracy.
Smaller h helps every method, but
RK4 gains far faster.

Order of accuracy: your first check on any result

If the global error is $O(h^p)$, then halving the step size divides the error by about 2^p :

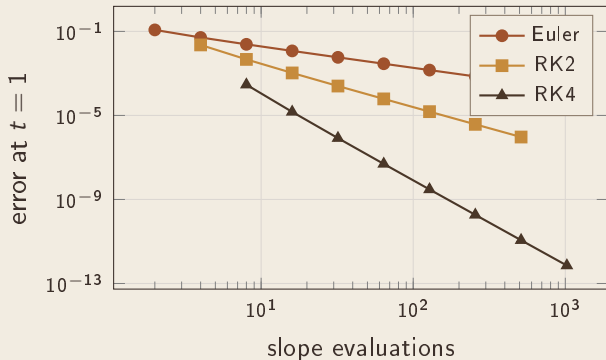
$$\frac{\text{error}(h)}{\text{error}(h/2)} \approx 2^p$$

	Order p	Expected ratio	Measured ($h = 2^{-7} \rightarrow 2^{-8}$)
Euler	1	2	2.0
RK2	2	4	4.0
RK4	4	16	16.1

A habit to build

Never trust a single run. Repeat with $h/2$: if the ratio is not near 2^p , something is wrong.

Accuracy per unit of work



RK4 costs four slope evaluations per step, but it takes far fewer steps.

- RK4 with 128 evaluations:
error $\approx 3 \times 10^{-9}$
- Euler with 256 evaluations:
error $\approx 7 \times 10^{-4}$

Compare at equal work

The right question is not “cost per step” but “cost for a given accuracy”.

When RK4 is not enough

Unknown dynamics

You do not know how small h must be. **Adaptive methods** (e.g. RK45) estimate the error at each step and adjust h automatically.

Stiff problems

Processes act on very different time scales, and an explicit method must take tiny steps to stay stable. **Implicit methods** (backward Euler, BDF) remain stable with much larger steps.

In practice

Library solvers such as `scipy.integrate.solve_ivp` offer these methods. Choose by the **problem**, not by habit — and always run the $h/2$ check.

Newton–Raphson: solving $f(x) = 0$

Start from a guess x_n . Replace f by its tangent line at x_n and find where the tangent crosses zero:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Repeat until

$$|x_{n+1} - x_n| < \varepsilon.$$

Example: Kepler's equation

For an elliptical orbit

$$f(E) = E - e \sin E - M,$$

$$f'(E) = 1 - e \cos E.$$

Given M and e , find E .

Newton–Raphson in action

Kepler's equation with $M = 1$, starting from $E_0 = M$. Error $|E_n - E_{\text{exact}}|$:

Iteration	$e = 0.5$	$e = 0.9$
1	8×10^{-2}	6×10^{-1}
2	2×10^{-3}	7×10^{-2}
3	6×10^{-7}	2×10^{-3}
4	9×10^{-14}	1×10^{-6}
5	—	4×10^{-13}

Near the root the number of correct digits roughly **doubles** each iteration (quadratic convergence).

It can fail

A poor guess, $f'(x) \approx 0$ or several roots can send it to the wrong root or to infinity. Check that $|f(x)|$ is small at the answer.

Which method for which problem?

Your problem	First choice
Smooth ODE, moderate accuracy	RK4, with an $h/2$ check
Unknown time scales, or you need error control	Adaptive RK (e.g. RK45)
Very different time scales (stiff)	Implicit / stiff solver
Nonlinear algebraic equation	Newton–Raphson
Learning how a method works	Euler

Whatever you choose: halve h and compare.

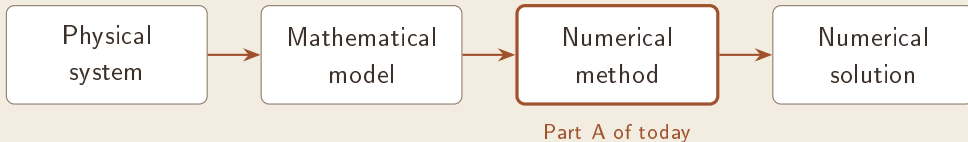
PART B

What kind of system is it?

Before running a simulation, or analysing data, ask what you should expect.

Isolated, closed or open · linear or nonlinear · deterministic or stochastic

From physical system to numerical solution



The methods you have just reviewed sit in the third box. Their **performance depends on the first two**.

The idea of Part B

Classify the system **before** you simulate. The classification tells you what the solution should look like, what to check, and which method from Part A is appropriate.

Three questions to ask of any system

Question	Classification	Why it matters for computing
What crosses the system boundary?	Isolated / Closed / Open	Which conservation laws to use and check
How do the variables interact?	Linear / Nonlinear	Solution behaviour; whether an exact solution exists; stiffness
How does it evolve, and where does randomness enter?	Deterministic / Stochastic	One trajectory, or an ensemble of them

These are **independent** labels for the same physical system. We take them in turn.

1. Classification by system boundary

System	Mass	Energy	Example	Governing balance
Isolated	no	no	Two-body orbit	E , momentum conserved
Closed	no	yes	Piston–cylinder	Energy balance
Open	yes	yes	Stirred reactor (CSTR)	Mass and energy balances

Why this is the first question

The boundary decides which equations you need, and it gives you quantities to **check afterwards**: conserved quantities for an isolated system, a balance for the others.

Isolated system: the two-body orbit

Point masses, Newtonian gravity, no external forces.

With $\mu = G(m_1 + m_2)$ and $r = \sqrt{x^2 + y^2}$:

$$\begin{aligned}\dot{x} &= v_x, & \dot{y} &= v_y, \\ \dot{v}_x &= -\frac{\mu x}{r^3}, & \dot{v}_y &= -\frac{\mu y}{r^3}\end{aligned}$$

Start from $x_0 = r_0$, $y_0 = 0$, $v_{x0} = 0$, $v_{y0} = v_0$. The value of v_0 decides circular, elliptical or escape orbit.

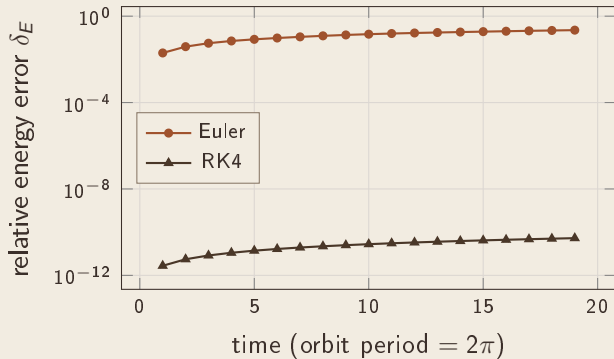
Hypothesis before simulating

Nothing crosses the boundary,
so

$$\epsilon = \frac{1}{2}v^2 - \frac{\mu}{r}, \quad h = xv_y - yv_x$$

should stay **constant**.

Does the numerical solution respect the physics?



Circular orbit, $\mu = 1$, $r_0 = 1$, $v_0 = 1$,
 $h = 0.01$, about three orbits.

$$\delta_E(t) = \frac{|\epsilon(t) - \epsilon(0)|}{|\epsilon(0)|}$$

Lesson

Euler's energy error grows to over 20% in three orbits; RK4's stays near 10^{-10} . A plausible-looking trajectory is **not** evidence of accuracy. Always monitor the conserved quantity.

Closed system: piston–cylinder

A gas in a piston–cylinder exchanges **energy but not mass** with its surroundings:

$$\boxed{dU = \delta Q - \delta W}, \quad \delta W = P dV, \quad PV = nRT.$$

What to expect

No conservation of mechanical energy. U changes only through heat and work across the boundary, so the **energy balance** is your check.

Methods from Part A

Coupled ODEs for the state: RK4. If the equation of state must be solved for a variable at each step: Newton–Raphson.

Open system: a stirred reactor (CSTR)¹

Reaction $A \rightarrow B$, first-order kinetics, flow Q through volume V :

$$\frac{dC_A}{dt} = \frac{Q}{V} (C_{A,\text{in}} - C_A) - kC_A$$

Hypothesis: set $dC_A/dt = 0$

$$C_{\text{ss}} = \frac{(Q/V) C_{A,\text{in}}}{Q/V + k}, \quad \tau = \frac{1}{Q/V + k}$$

C_A relaxes exponentially to C_{ss} with time constant τ .

A free test

This ODE is **linear**, so the exact solution is known: compare Euler and RK4 against it, and repeat the $h/2$ check.

¹A model of a stirred reactor – most commonly called a Continuous Stirred-Tank Reactor (CSTR) – uses mass and energy balances to describe how chemical concentration and temperature change inside a mixed vessel over time

2. Mathematical behaviour: linear systems

Mass-spring-damper, $m\ddot{x} + c\dot{x} + kx = F(t)$. With $x_1 = x$, $x_2 = \dot{x}$:

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = \frac{1}{m} [F(t) - cx_2 - kx_1].$$

What to expect

Superposition holds. With $c > 0$ free motion decays; if $c^2 < 4mk$ it oscillates at $\omega_d = \sqrt{k/m - (c/2m)^2}$. Periodic forcing gives a response at the forcing frequency.

How to check

Compare with the analytic solution, and read behaviour off the eigenvalues. RK4 is usually a safe default.

Nonlinear systems: the Van der Pol oscillator

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$$

What to expect

For $|x| < 1$ the damping is negative and pumps energy in; for $|x| > 1$ it removes energy. The result is a **limit cycle**: a self-sustained oscillation to which trajectories converge from different initial conditions. Superposition fails.

Methods from Part A

For large μ the motion has fast and slow phases: expect **stiffness**. Use an adaptive or implicit solver, and always test several initial conditions.

Linear versus nonlinear

	Linear	Nonlinear
Superposition	Yes	Generally no
Exact solution	Often available	Often not
Typical behaviour	Decay, growth, response at forcing frequency	Limit cycles, several steady states, chaos
Numerical method	RK4 usually fine	May need adaptive or stiff solvers
Example	Mass–spring–damper	Van der Pol

The mathematical structure influences which method from Part A you should trust.

Conservative versus dissipative

Conservative

Energy is constant (two-body orbit).

Any drift you see is numerical: suspect the step size or the method.

Dissipative

Energy decreases (damped oscillator).

Decay is physical; a **rise** would be a warning sign. Driven systems such as Van der Pol can sustain oscillation.

Know which kind of system you have before you interpret the energy curve.

3. Time variability, uncertainty, stochasticity

How does the system change with time, and how do we represent what we do not know?

Time variability The state changes with time, $x(t)$. Fully compatible with deterministic dynamics.

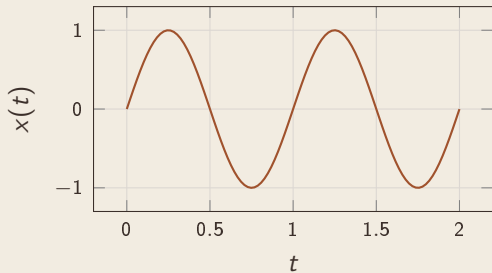
Uncertainty We do not know inputs, parameters or measurements exactly.

Stochasticity Randomness is part of the **dynamics**.

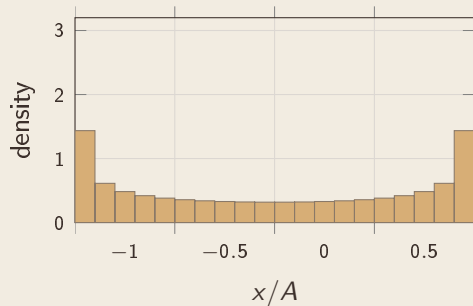
Related, but **not synonyms**.

Randomness is often just time variability

With time: $x = A \sin \omega t$



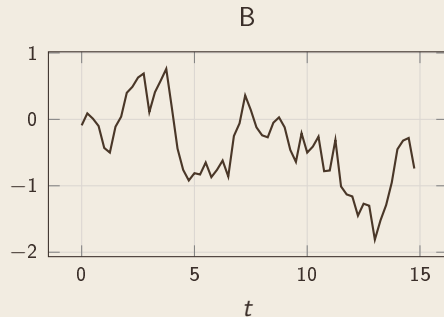
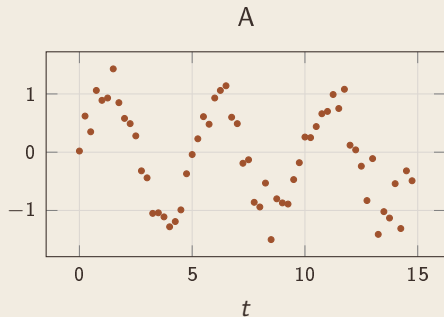
Time discarded: histogram



Same values. Without time they have variance $A^2/2 > 0$ and look like a spread of random numbers.

$$\text{Var}(X) > 0 \not\Rightarrow \text{stochastic process}$$

Which one is random?



A is a deterministic oscillation plus measurement noise, $Y_i = X(t_i) + \varepsilon_i$. B is a random walk: randomness is in the **dynamics**. A histogram discards the **time ordering** that distinguishes them.

Separating dynamics from uncertainty

1. Identify temporal structure (trend, periodicity, transients).
2. Fit a dynamic model $\hat{X}(t)$.
3. Analyse the residuals

$$e_i = Y_i - \hat{X}(t_i).$$

Structured residuals

Systematic temporal pattern: the model is probably **missing dynamics**.

Noise-like residuals

Unexplained variation may be measurement error, parameter uncertainty or stochastic forcing. It is not automatically “random”.

Uncertain inputs, deterministic dynamics

$$\dot{x} = f(x, \theta, t), \quad \theta \sim p(\theta).$$

Monte Carlo: draw $\theta^{(1)}, \dots, \theta^{(N)}$, run the deterministic solver (e.g. RK4) for each, and study the ensemble:

$$\widehat{\mathbb{E}}[X(t)], \quad \widehat{\text{Var}}[X(t)].$$

Example

Orbit with uncertain initial speed $v_0 \pm \Delta v$: each run is a deterministic orbit, together they show how the uncertainty spreads.

Many deterministic simulations $\rightarrow$ a statistical description

Stochastic dynamics: Euler–Maruyama

$$dX = f(X, t) dt + g(X, t) dW_t, \quad dW_t \sim \mathcal{N}(0, dt).$$

The Euler–Maruyama update is Euler’s method plus a random kick:

$$X_{n+1} = X_n + f(X_n, t_n) \Delta t + g(X_n, t_n) \sqrt{\Delta t} \xi_n, \quad \xi_n \sim \mathcal{N}(0, 1).$$

Link to Part A

With $g = 0$ it reduces to Euler. Even with identical initial conditions, repeated runs give **different** trajectories. Its strong convergence order is only $1/2$.

One simulation versus an ensemble

A deterministic run gives $x(t)$. A stochastic run gives one *realisation* $X^{(i)}(t)$. Repeat N times:

$$\widehat{\mathbb{E}}[X(t)] = \frac{1}{N} \sum_{i=1}^N X^{(i)}(t), \quad \widehat{\text{Var}}[X(t)] = \frac{1}{N-1} \sum_{i=1}^N \left(X^{(i)} - \widehat{\mathbb{E}} \right)^2.$$

The question changes

From “what trajectory occurs?” to “what is the distribution of trajectories?” The sampling error of the ensemble mean shrinks only as $1/\sqrt{N}$.

Numerical error is not physical uncertainty

Physical / modelling	Numerical
Measurement uncertainty	Discretisation error (finite h)
Parameter uncertainty	Round-off error (finite precision)
Initial-condition uncertainty	
Model uncertainty (neglected physics)	

Connection to Part A

Halving h reduces only the discretisation error. More numerical precision does **not** give more physical certainty.

One system, several labels

The labels are independent. A stirred reactor can be:

Dimension	CSTR example
Boundary	Open: mass crosses it
Mathematics	Linear (first-order kinetics) or nonlinear (other kinetics)
Time dependence	Dynamic: concentration changes with time
Variability	Deterministic, unless random forcing is modelled

No contradiction: each label answers a different question.

A note on quantum systems

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

The labels again

For an isolated system the Schrödinger evolution of ψ is linear and deterministic. Measurement outcomes are probabilistic, with probabilities given by $|\psi|^2$.

The same habit

What should be conserved? Here the norm, $\int |\psi|^2 dx = 1$. Check that your numerical solution preserves it, just as you checked the energy.

Linear, deterministic dynamics with a probabilistic description of outcomes: one more reminder that the labels are independent.

What should I expect, and how do I check?

System type	What to expect	Check with
Isolated, conservative	Constant ϵ , h , momentum	δ_E , δ_h against h and method
Closed	Energy changes only via Q , W	Energy balance
Open	Steady state, sustained oscillation, or other driven behaviour	Steady state by root finding; time constant
Linear, dissipative	Decay; superposition	Exact solution; eigenvalues
Nonlinear	Limit cycles, several states, possible stiffness	Vary initial conditions; adaptive or stiff solver
Uncertain or stochastic	Spread of outcomes	Ensemble statistics; behaviour with N
Quantum (isolated)	Constant norm; probabilistic outcomes	$\int \psi ^2 dx = 1$

Verification and validation

Verification

Did we solve the model correctly?

- $h/2$ test and order of accuracy
- conserved quantities
- exact solutions where they exist

Validation

Does the model represent the system?

- comparison with experimental data
- prediction of independent observations
- examining the modelling assumptions

Part A gave you the verification tools; Part B tells you what to compare with.

Before you simulate: five questions

1. What crosses the boundary? (*which balance or conserved quantity*)
2. Linear or nonlinear? (*exact solution? stiffness?*)
3. Conservative or dissipative? (*what should the energy do?*)
4. Where, if anywhere, does randomness enter? (*inputs, measurements, dynamics*)
5. What do I expect, and how will I check it?

Then choose the method (Part A), and halve h to verify.

TO TAKE AWAY

Hypothesis first, numerical methods second.

Know the system, expect its behaviour, choose the method, check the result.